

Assignment 3

- a. Based on accuracy which model is the best one?

The best model is LDA with accuracy = 98%.

- b. For each of the 11 other models, explain why you think it does not perform as well as the best one

Naive Bayes (GaussianNB) - 95.3%: Assumes feature independence and Gaussian distributions. This limits performance.

Linear Regression (rounded) - 97.3%: Regression isn't designed for classification. Rounding works decently, but the model isn't optimized for class separation.

Polynomial Regression (degree 2) - 95.3%: Adds nonlinearity, but rounding predictions still introduces errors. Overfitting noise without improving class separation.

Polynomial Regression (degree 3) - 86.7%: Too much complexity strong overfitting and poor generalization. Struggles with separating the classes.

k-NN - 94.0%: Sensitive to choice of k and distance metric. Noise and overlapping class boundaries in Iris reduce accuracy.

QDA - 97.3%: Strong model, but quadratic boundaries can overfit when class covariance estimates are slightly noisy, leading to a few more errors.

SVM (LinearSVC) - 94.0%: Linear SVM is less flexible. Misclassifies more points around overlapping class boundaries.

Decision Tree - 95.3%: Trees tend to overfit small datasets.

Random Forest - 96.0%: Reduces overfitting, but still not as efficient as LDA for a simple, low-dimensional datasets.

Extra Trees - 96.0%: Similar to Random Forest, but more randomized splits. Still slightly worse than LDA.

Neural Network (MLP) - 97.3%: Very strong, but with small datasets, NNs may slightly overfit or fail to capture linear separability as cleanly as LDA

CompareMLModelsV2: 2-fold cross-validated predictions on Iris (150 samples)

=== Model: Naive Bayes (GaussianNB) ===

Confusion Matrix (rows: true classes 0,1,2 ; cols: predicted 0,1,2):

```
[[50 0 0]
 [ 0 46 4]
 [ 0 3 47]]
```

Sum of confusion matrix entries (should be 150): 150

Accuracy: 0.9533

=== Model: Linear Regression (LR → round) ===

Confusion Matrix (rows: true classes 0,1,2 ; cols: predicted 0,1,2):

```
[[50 0 0]
 [ 0 48 2]
 [ 0 2 48]]
```

Sum of confusion matrix entries (should be 150): 150

Accuracy: 0.9733

=== Model: Polynomial deg2 Regression (LR poly2 → round) ===

Confusion Matrix (rows: true classes 0,1,2 ; cols: predicted 0,1,2):

```
[[49 1 0]
 [ 0 48 2]
 [ 0 4 46]]
```

Sum of confusion matrix entries (should be 150): 150

Accuracy: 0.9533

=== Model: Polynomial deg3 Regression (LR poly3 → round) ===

Confusion Matrix (rows: true classes 0,1,2 ; cols: predicted 0,1,2):

```
[[48 2 0]
 [ 0 44 6]
 [ 2 10 38]]
```

Sum of confusion matrix entries (should be 150): 150

Accuracy: 0.8667

=== Model: k-NN (KNeighborsClassifier) ===

Confusion Matrix (rows: true classes 0,1,2 ; cols: predicted 0,1,2):

```
[[50 0 0]
```

```
 [ 0 45 5]
```

```
 [ 0 4 46]]
```

Sum of confusion matrix entries (should be 150): 150

Accuracy: 0.9400

=== Model: LDA (LinearDiscriminantAnalysis) ===

Confusion Matrix (rows: true classes 0,1,2 ; cols: predicted 0,1,2):

```
[[50 0 0]
```

```
 [ 0 48 2]
```

```
 [ 0 1 49]]
```

Sum of confusion matrix entries (should be 150): 150

Accuracy: 0.9800

=== Model: QDA (QuadraticDiscriminantAnalysis) ===

Confusion Matrix (rows: true classes 0,1,2 ; cols: predicted 0,1,2):

```
[[50 0 0]
```

```
 [ 0 47 3]
```

```
 [ 0 1 49]]
```

Sum of confusion matrix entries (should be 150): 150

Accuracy: 0.9733

=== Model: SVM (LinearSVC) ===

Confusion Matrix (rows: true classes 0,1,2 ; cols: predicted 0,1,2):

```
[[49 1 0]
```

```
 [ 1 45 4]
```

```
 [ 0 3 47]]
```

Sum of confusion matrix entries (should be 150): 150

Accuracy: 0.9400

=== Model: Decision Tree (DecisionTreeClassifier) ===

Confusion Matrix (rows: true classes 0,1,2 ; cols: predicted 0,1,2):

```
[[50 0 0]
```

```
 [ 0 46 4]
```

```
[ 0 3 47]]
```

Sum of confusion matrix entries (should be 150): 150

Accuracy: 0.9533

=== Model: Random Forest (RandomForestClassifier) ===

Confusion Matrix (rows: true classes 0,1,2 ; cols: predicted 0,1,2):

```
[[50 0 0]
```

```
[ 0 47 3]
```

```
[ 0 3 47]]
```

Sum of confusion matrix entries (should be 150): 150

Accuracy: 0.9600

=== Model: ExtraTrees (ExtraTreesClassifier) ===

Confusion Matrix (rows: true classes 0,1,2 ; cols: predicted 0,1,2):

```
[[50 0 0]
```

```
[ 0 46 4]
```

```
[ 0 2 48]]
```

Sum of confusion matrix entries (should be 150): 150

Accuracy: 0.9600

=== Model: Neural Network (MLPClassifier) ===

Confusion Matrix (rows: true classes 0,1,2 ; cols: predicted 0,1,2):

```
[[49 1 0]
```

```
[ 0 48 2]
```

```
[ 0 1 49]]
```

Sum of confusion matrix entries (should be 150): 150

Accuracy: 0.9733

a. Does the program use k-fold cross-validation?

No, it does not use k-fold CV

b. What percentage of the data set was used to train the DBN model?

by default, 75% is for training, 25% is for testing.

c. How many samples are in the test set?

$25\% * 1797 \approx 450$

d. How many samples are in the training set?

$75\% * 1797 \approx 1347$

e. How many features are in test set?

Each sample has 64 features, so test set shape = (450, 64)

f. How many features are in the training set?

64 features

g. How many classes are there?

10 classes

h. List the classes.

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

[START] Pre-training step:

```
>> Epoch 1 finished  RBM Reconstruction error 2.740666
>> Epoch 2 finished  RBM Reconstruction error 1.805594
>> Epoch 3 finished  RBM Reconstruction error 1.432523
>> Epoch 4 finished  RBM Reconstruction error 1.274614
>> Epoch 5 finished  RBM Reconstruction error 1.127715
>> Epoch 6 finished  RBM Reconstruction error 1.016233
>> Epoch 7 finished  RBM Reconstruction error 0.931797
>> Epoch 8 finished  RBM Reconstruction error 0.904623
>> Epoch 9 finished  RBM Reconstruction error 0.891258
>> Epoch 10 finished  RBM Reconstruction error 0.790757
>> Epoch 1 finished  RBM Reconstruction error 2.880171
>> Epoch 2 finished  RBM Reconstruction error 1.491411
>> Epoch 3 finished  RBM Reconstruction error 1.120919
>> Epoch 4 finished  RBM Reconstruction error 0.921536
>> Epoch 5 finished  RBM Reconstruction error 0.763747
>> Epoch 6 finished  RBM Reconstruction error 0.667087
>> Epoch 7 finished  RBM Reconstruction error 0.573752
>> Epoch 8 finished  RBM Reconstruction error 0.529821
```

```
>> Epoch 9 finished  RBM Reconstruction error 0.480695
>> Epoch 10 finished  RBM Reconstruction error 0.456472
[END] Pre-training step
[START] Fine tuning step:
>> Epoch 1 finished  ANN training loss 19.815676
>> Epoch 2 finished  ANN training loss 13.012750
>> Epoch 3 finished  ANN training loss 9.025525
>> Epoch 4 finished  ANN training loss 7.295055
>> Epoch 5 finished  ANN training loss 6.229197
>> Epoch 6 finished  ANN training loss 5.808220
>> Epoch 7 finished  ANN training loss 5.241246
>> Epoch 8 finished  ANN training loss 4.761866
>> Epoch 9 finished  ANN training loss 4.898449
>> Epoch 10 finished  ANN training loss 4.430443
>> Epoch 11 finished  ANN training loss 4.188848
>> Epoch 12 finished  ANN training loss 4.099041
>> Epoch 13 finished  ANN training loss 4.149002
>> Epoch 14 finished  ANN training loss 4.126827
>> Epoch 15 finished  ANN training loss 4.024709
>> Epoch 16 finished  ANN training loss 3.837017
>> Epoch 17 finished  ANN training loss 3.789805
>> Epoch 18 finished  ANN training loss 3.749225
>> Epoch 19 finished  ANN training loss 3.768161
>> Epoch 20 finished  ANN training loss 3.448430
>> Epoch 21 finished  ANN training loss 3.973341
>> Epoch 22 finished  ANN training loss 3.486660
>> Epoch 23 finished  ANN training loss 3.562496
>> Epoch 24 finished  ANN training loss 3.220933
>> Epoch 25 finished  ANN training loss 3.473933
>> Epoch 26 finished  ANN training loss 3.096372
>> Epoch 27 finished  ANN training loss 3.127187
>> Epoch 28 finished  ANN training loss 3.654857
>> Epoch 29 finished  ANN training loss 3.176199
>> Epoch 30 finished  ANN training loss 3.204959
>> Epoch 31 finished  ANN training loss 3.372546
>> Epoch 32 finished  ANN training loss 3.147058
```

```
>> Epoch 33 finished ANN training loss 3.235672
>> Epoch 34 finished ANN training loss 2.879812
>> Epoch 35 finished ANN training loss 2.878245
>> Epoch 36 finished ANN training loss 3.084891
>> Epoch 37 finished ANN training loss 3.132054
>> Epoch 38 finished ANN training loss 2.776393
>> Epoch 39 finished ANN training loss 3.132948
>> Epoch 40 finished ANN training loss 2.989652
>> Epoch 41 finished ANN training loss 2.867335
>> Epoch 42 finished ANN training loss 2.809363
>> Epoch 43 finished ANN training loss 2.884767
>> Epoch 44 finished ANN training loss 3.035932
>> Epoch 45 finished ANN training loss 2.848164
>> Epoch 46 finished ANN training loss 2.913252
>> Epoch 47 finished ANN training loss 2.813828
>> Epoch 48 finished ANN training loss 2.736477
>> Epoch 49 finished ANN training loss 2.766231
>> Epoch 50 finished ANN training loss 2.839658
>> Epoch 51 finished ANN training loss 2.699174
>> Epoch 52 finished ANN training loss 2.548628
>> Epoch 53 finished ANN training loss 2.651098
>> Epoch 54 finished ANN training loss 2.472910
>> Epoch 55 finished ANN training loss 2.586327
>> Epoch 56 finished ANN training loss 2.511920
>> Epoch 57 finished ANN training loss 2.684920
>> Epoch 58 finished ANN training loss 2.370807
>> Epoch 59 finished ANN training loss 2.716494
>> Epoch 60 finished ANN training loss 2.562766
>> Epoch 61 finished ANN training loss 2.397317
>> Epoch 62 finished ANN training loss 2.461418
>> Epoch 63 finished ANN training loss 2.493879
>> Epoch 64 finished ANN training loss 2.541392
>> Epoch 65 finished ANN training loss 2.619213
>> Epoch 66 finished ANN training loss 2.511461
>> Epoch 67 finished ANN training loss 2.588723
>> Epoch 68 finished ANN training loss 2.558595
```

```
>> Epoch 69 finished ANN training loss 2.794101
>> Epoch 70 finished ANN training loss 2.243804
>> Epoch 71 finished ANN training loss 2.400391
>> Epoch 72 finished ANN training loss 2.415319
>> Epoch 73 finished ANN training loss 2.235596
>> Epoch 74 finished ANN training loss 2.320560
>> Epoch 75 finished ANN training loss 2.365965
>> Epoch 76 finished ANN training loss 2.373090
>> Epoch 77 finished ANN training loss 2.344638
>> Epoch 78 finished ANN training loss 2.370614
>> Epoch 79 finished ANN training loss 2.378325
>> Epoch 80 finished ANN training loss 2.296690
>> Epoch 81 finished ANN training loss 2.207147
>> Epoch 82 finished ANN training loss 2.213116
>> Epoch 83 finished ANN training loss 2.479594
>> Epoch 84 finished ANN training loss 2.184803
>> Epoch 85 finished ANN training loss 2.145601
>> Epoch 86 finished ANN training loss 2.331894
>> Epoch 87 finished ANN training loss 2.236025
>> Epoch 88 finished ANN training loss 2.343825
>> Epoch 89 finished ANN training loss 1.923268
>> Epoch 90 finished ANN training loss 2.162285
>> Epoch 91 finished ANN training loss 2.198585
>> Epoch 92 finished ANN training loss 2.443944
>> Epoch 93 finished ANN training loss 2.322807
>> Epoch 94 finished ANN training loss 2.092216
>> Epoch 95 finished ANN training loss 2.326477
>> Epoch 96 finished ANN training loss 2.240110
>> Epoch 97 finished ANN training loss 2.341988
>> Epoch 98 finished ANN training loss 1.917154
>> Epoch 99 finished ANN training loss 2.006803
>> Epoch 100 finished ANN training loss 1.919281
[END] Fine tuning step
Done.
Accuracy: 0.997222
```


